KANI'S LEMMA AND ITS APPLICATIONS TO ISOGENY-BASED CRYPTOGRAPHY

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What is an isogeny?

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SECTION 1.1

What is an isogeny?

For a mathematician.

ABELIAN VARIETIES

Definition [Mil08]

An **abelian variety** over a field $\Bbbk$ is a projective variety A together with morphisms

$$\begin{aligned} m &: A \times A \to A, \\ \iota &: A \to A \\ e &: \mathrm{Spec}(\Bbbk) \to A \end{aligned}$$

making A a commutative group object in $\mathsf{Var}_{\Bbbk}$.

Example

Every **elliptic curve** E is an abelian variety of dimension 1.

$$y^2 = x^3 + ax + b$$

ISOGENIES

Definition [Mil08, Prop. 7.1]

A homomorphism $\varphi : A \to B$ between abelian varieties is called an **isogeny** if

- φ is surjective,
- $\dim A = \dim B$.

The **degree** of φ is defined as:

$$\deg \varphi := [\Bbbk(A) : \Bbbk(B)]$$

Note that the degree is multiplicative.

Kernel $\longleftrightarrow$ Isogeny [Mum70, Ap. 6, Thm. 4]

Given an abelian variety A, then there is a bijection between the following two sets:

- finite subgroups $G \subseteq A(\bar{\Bbbk})$,
- separable isogenies $\varphi : A \to B$ with $\ker \varphi = G$.

PRINCIPALLY POLARIZED ABELIAN VARIETIES

Definition [HS00]

A **polarization** on A is an isogeny

$$\lambda : A \to \hat{A}$$

induced by an ample line bundle. It is **principal** if it is an isomorphism. This implies $\deg \lambda = 1$.

We call the tuple (A, λ) a **(principally) polarized abelian variety (p.p.a.v.)**.

Example [Sil09]

Every elliptic curve E carries a canonical principal polarization

$$\begin{aligned} \lambda_E : E &\to \hat{E} \\ P &\mapsto [P] - [O]. \end{aligned}$$

ISOGENIES OF P.P.A.V.S

Definition [Rob22]

An n-isogeny $\varphi : A \to B$ between p.p.a.v.s (A, λ_A) and (B, λ_B) is called **compatible** if

$$\varphi^* \lambda_B = [n]_A \circ \lambda_A = n \cdot \lambda_A,$$

where $\varphi^* \lambda_B := \hat{\varphi} \circ \lambda_B \circ \varphi$.

Illustration

$$\begin{tikzcd}
\hat{A} \arrow[d, shift left, "\lambda_A^{-1}"] & \hat{B} \arrow[l, "\hat{\varphi}"'] \arrow[d, shift left, "\lambda_B^{-1}"] \\
A \arrow[u, shift left, "\lambda_A"] \arrow[r, "\varphi"'] \arrow[loop, out=-90, in=180, distance=8em, "{[n]_A}"] & B \arrow[u, shift left, "\lambda_B"]
\end{tikzcd}$$

Remark

Compatibility means the diagram above **commutes**.

1.2

SECTION 1.2

What is an isogeny?

For a computer-scientist.

REPRESENTING ISOGENIES

Why Representations?

Existence results are not enough for cryptographic purposes — we need **explicit, efficient representations** to:

- store φ,
- evaluate φ on points $P \in A(\mathbb{F}_{q^k})$,
- carry out computations efficiently.

Definition (Efficient Representation)[Ler22]

A representation of $\varphi : A \to B$ of degree d over $\mathbb{F}_q$ consists of:

- a datum $D \in \{0, 1\}^*$,
- an algorithm $\mathcal{A}(D, P) \mapsto \varphi(P)$,

We say it is *efficient* if:

- $|D| = \mathrm{poly}(\log d, \log q)$,
- $\mathcal{A}$ runs in $\mathrm{poly}(\log d, k \cdot \log q)$.

VÉLU'S FORMULAS

Theorem (Vélu) [Coh+05]

Let E be an elliptic curve and $G \subseteq E(\bar{\mathbb{k}})$ a finite subgroup. Then there exists an elliptic curve $E' = {}^{E}\!/_{G}$ and a separable isogeny

$$\varphi_G : E \to {}^{E}\!/_{G}$$

with $\ker \varphi_G = G$, computable explicitly in $\mathcal{O}(|G|)$ operations.

Smoothness

For $\deg \varphi = \prod l_i^{e_i}$ smooth, we decompose $\varphi = \varphi_k \circ \cdots \circ \varphi_1$ with $\deg \varphi_i = l_i$ — each factor computable efficiently.

Higher Dimensions [Rob22]

For p.p.a.v.s of dimension g, analogous **Vélu-like formulas** exist using theta models — the higher-dimensional analogue of Weierstrass models.

KERNEL REPRESENTATION — THE STANDARD APPROACH

The Big Picture

For a separable isogeny $\varphi : A \to B$, giving

$$\ker \varphi = G \subseteq A(\bar{k})$$

together with generators of G yields an **efficient representation** of φ — *provided* $\deg \varphi$ *is smooth.*

Looking ahead

What if $\deg \varphi$ is **not** smooth?
$\longrightarrow$ HD-Representation.

Consequence

- **Space-efficient:** kernel generators have size $\mathcal{O}(\log \deg \varphi)$.
- **Time-efficient:** evaluation runs in $\mathrm{poly}(\log \deg \varphi, k \log q)$.
- **Limitation:** breaks down for *non-smooth* degrees.

SECTION 2.

Kani's Lemma and Zarhin's trick.

2.1

SECTION 2.1

Kani's Lemma and Zarhin's trick.

From the theoretical site.

ISOGENY FACTORISATION CONFIGURATION

Definition [Kan97]

Let (A, λ_A), (B, λ_B) be p.p.a.v.s and $n \geq 2$. A **factorisation configuration of order** n is a triple (φ, H_1, H_2) with:

- $\varphi : A \to B$ an isogeny,
- $H_1, H_2 \subseteq \ker \varphi$ subgroups,
- $\mathrm{ord}(H_1) + \mathrm{ord}(H_2) = n$,
- $\mathrm{ord}(H_1) \cdot \mathrm{ord}(H_2) = \deg \varphi$.

If $H_1 \cap H_2 = \{0\}$, it is called an **isogeny diamond configuration**.

$$\begin{tikzcd} A \arrow[r, "\varphi_1"] \arrow[d, "\varphi_2"'] \arrow[dr, "\varphi"] & A_1 \arrow[d, "\psi_2"] \\ A_2 \arrow[r, "\psi_1"'] & B \end{tikzcd}$$

In words

Two different factorizations of φ via intermediate varieties A_1, A_2.

Theorem (Kani's Lemma)

Given an isogeny factorization configuration of order n $(\varphi, \varphi_1, \varphi_1', \varphi_2, \varphi_2')$, where φ_1 is a d_1-isogeny and φ_2 is a d_2-isogeny. Then

$$F := \begin{pmatrix} \varphi_1 & -\tilde{\varphi}_1' \\ \varphi_2 & \tilde{\varphi}_2' \end{pmatrix} : A \times B \to A_1 \times A_2$$

is an $n = (d_1 + d_2)$-isogeny between the p.p.a.v.'s with their product polarizations.

Key Insight

If d_1 and d_2 are coprime, then:

$$\ker(F) = \{([d_1](P), -\varphi(P)) \mid P \in A[n]\}.$$

Isogeny Diamond

$$\begin{tikzcd} & A \arrow[dl, "\varphi_1"'] \arrow[dd, "\varphi"] \arrow[dr, "\varphi_2"] & \\ A_1 \arrow[dr, "\varphi_1'"'] & & A_2 \arrow[dl, "\varphi_2'"] \\ & B & \end{tikzcd}$$

BUILDING AN ISOGENY DIAMOND

Zarhin's trick

Given any $m > 0$, write $m = a^2 + b^2 + c^2 + d^2$ by Lagrange. Zarhin's Trick gives us m-isogenies

$$\alpha_A : A^u \to A^u,$$
$$\alpha_B : B^u \to B^u,$$

with $u \in \{1, 2, 4\}$ via scalar multiplications.

Diamond condition

Since α consists only of scalar multiplications, we immediately have

$$\alpha_A \circ \varphi^u = \varphi^u \circ \alpha_B.$$

The resulting diamond

$$\begin{tikzcd}
 & A^u \arrow[dl, "\alpha_A"'] \arrow[dr, "\varphi^u"] & \\
A^u \arrow[dr, "\varphi^u"'] & & B^u \arrow[dl, "\alpha_B"] \\
 & B^u &
\end{tikzcd}$$

Lemma 3.4 (Embedding Lemma)

For any $m > 0$, an n-isogeny $\varphi : A \to B$ of p.p.a.v.'s of dimension g can be embedded into an $(n+m)$-isogeny F of dimension $2 \cdot u \cdot g$, with $u \in \{1, 2, 4\}$. Concretely:

$$F := \begin{pmatrix} \alpha_A & -\tilde{\varphi} \\ \varphi & \tilde{\alpha}_B \end{pmatrix} : A^u \times B^u \to A^u \times B^u.$$

Key consequence

We can recover φ from F via

$$A \xrightarrow{\iota} A^u \times B^u \xrightarrow{F} A^u \times B^u \xrightarrow{\pi} B.$$

Every isogeny can be embedded into a higher-dimensional isogeny of **smooth degree**.

Construction

$$\begin{tikzcd} A^u \arrow[r, "\alpha_A"] \arrow[d, "\varphi^u"'] & A^u \arrow[d, "\varphi^u"] \\ B^u \arrow[r, "\alpha_B"'] & B^u \end{tikzcd}$$

HD-REPRESENTATION

Definition (HD-Representation)

Any representation of F_φ from the Embedding Lemma is called an **HD-representation** of φ.

Theorem [Rob23]

Let $\varphi : A \to B$ be an n-isogeny and $N = \prod l_i^{e_i} > n$ coprime to n. Given evaluations of φ on $A[N]$, we can represent φ as a smooth-degree $(2ug)$-dimensional N-isogeny F_φ.

Strategy

- Choose N smooth, $\gcd(N, n) = 1$
- Embed φ into F_φ via Embedding Lemma
- Recover φ via $\varphi = \pi \circ F_\varphi \circ \iota$

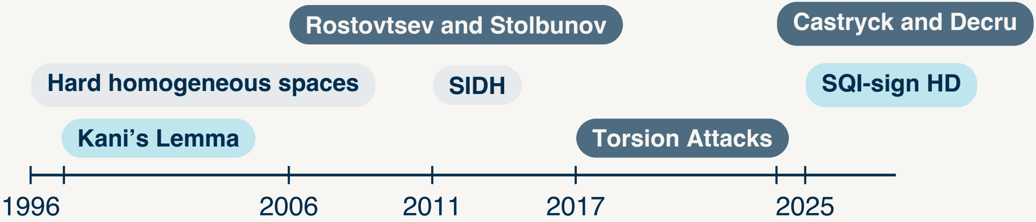
Key takeaway

Every isogeny — regardless of its degree — admits an efficient representation.

SECTION 3.

A little History.

TIMELINE


Rostovtsev and Stolbunov
Castryck and Decru
Hard homogeneous spaces
SIDH
SQI-sign HD
Kani’s Lemma
Torsion Attacks
1996
2006
2011
2017
2025

SECTION 4.

A new dimension.

4.1

SECTION 4.1

A new dimension.

Destructive Applications.

SIDH

Setup

Prime $p = f \cdot l_A^{e_A} \cdot l_B^{e_B} \pm 1$, supersingular curve $E_0/\mathbb{F}_{p^2}$ with public torsion bases

$$\langle P_A, Q_A \rangle = E_0[\,l_A^{e_A}\,],$$
$$\langle P_B, Q_B \rangle = E_0[\,l_B^{e_B}\,].$$

Key Exchange

Alice computes secret $\varphi_A : E_0 \to E_A$, and also publishes $\varphi_A(P_B), \varphi_A(Q_B)$.
Bob does the same.
Shared secret: $j(E_{AB}) = j(E_{BA})$.

The SIDH diagram

$$\begin{tikzcd} E_0 \arrow[r, "\varphi_A"] \arrow[d, "\varphi_B"'] & E_A \arrow[d, "\varphi'_B"] \\ E_B \arrow[r, "\varphi'_A"'] & E_{AB} \end{tikzcd}$$

Fatal extra information

Public keys reveal the action of φ_A on $E_0[\,l_B^{e_B}\,]$ and vice versa — exactly the torsion information needed for the HD-representation.

ATTACK ON SIDH

Setup

W.l.o.g. assume $l_A^{e_A} > l_B^{e_B}$. Set $n := l_A^{e_A} - l_B^{e_B}$, then $l_A^{e_A}, l_B^{e_B}, n$ are pairwise coprime. Construct isogeny with Zarhin's trick:

$$\alpha_0 : E_0^u \to E_0^u,$$
$$\alpha_B : E_B^u \to E_B^u.$$

Isogeny Diamond

$$\begin{tikzcd}
 & E_0^u \arrow[dl, "\alpha_0"'] \arrow[dr, "\varphi_B^u"] & \\
E_0^u \arrow[dr, "\varphi_B^u"'] & & E_B^u \arrow[dl, "\alpha_B"] \\
 & E_B^u &
\end{tikzcd}$$

Key Recovery

Bob's reveals φ_B on $E_0[\,l_A^{e_A}\,]$

$\Rightarrow$ Bob's secret isogeny is recovered in **polynomial time**.

4.2

SECTION 4.2

A new dimension.

Constructive Applications.

CONSTRUCTIVE APPLICATIONS

A new dimension

The HD-representation enables isogenies of **arbitrary degree** — not just smooth. This unlocks a new approach to isogeny-based signature schemes.

Key insight

Higher-dimensional isogenies shift computational cost and relax constraints on the underlying prime p — making higher security levels feasible.

Higher dimensional schemes

- **SQIsignHD [Dar+24]**
  dimension 4/8, non-smooth response isogeny
- **SQIPrime [DF25]**
  dimension 2, non-smooth *challenge* isogeny, no smooth degrees anywhere
- **PRISM [Bas+25]**
  large prime degree isogenies

SECTION 5.

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